

On Externalization

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Symmetry (Breaking) in Syntax
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0. An Asymmetric Digression on Symmetry (Breaking)

a. Andrea's Dynamic Antisymmetry (Moro (2000)) is intrinsically derivational. The LCA is not.

b. The illicit merger of two heads can be neutralized by moving one (Andrea) or by recourse to singleton set formation, as in my (2008a) "Antisymmetry and the Lexicon", which also leads to the following (vs. Chomsky's (1970) "Remarks on Nominalization"), in combination with unaccusativity and with Hale & Keyser (1993; 2002):

c. Nouns never have a complement and never project; Verbs always have a complement and always fully project. (All verbs are light verbs. What are light nouns?) The noun-verb distinction is induced by antisymmetry. With no space for any other category on the same plane, which leads to:

d. All other candidates for categoryhood (e.g. complementizer, adposition, affixes) must be either i) a subtype of N, or ii) a subtype of V, or iii) complex.

e. There are in the syntax semantically contentful elements with no phonetic realization (silent elements of various kinds). Are there in the syntax phonologically or phonetically contentful elements with no semantic realization, i.e. are there true expletives? - If not, why not? If so, why so?

f. Suppletion is usually thought of in terms of distinct phonological realizations of one syntactic element. Alternatively,¹ suppletion is better thought of as involving two related syntactic elements, each of which gives rise to separate questions, if we are sufficiently ambitious/curious.

g. Antihomophony says that, within the functional part of the lexicon of a given language, phonological identity implies identity.² Antisynonymy says that phonological non-identity implies non-synonymy.³ (Maurice Gross had noted a long time ago that no two French verbs have exactly the same syntactic properties.) As if perfect synonyms would be an instance of two (or more) phonologies intolerably 'symmetric' to a given interpretation and perfect homonyms an instance of two (or more) interpretations intolerably 'symmetric' to a given phonology.

h. Negative and positive integers could be thought of as symmetric to one another. Could/Should negative and positive polarity values be thought of in the same

¹Cf. Kayne (2019; to appear).

²On antihomophony, cf. Kayne (2026) and references cited there.

³On antisynonymy, cf. Kayne (to appear).

way? It seems not. After all, negative sentences are generally formed by the addition of some negative morpheme to a positive sentence. Whereas no language forms ordinary positive/affirmative sentences by adding a positive/affirmative morpheme to a negative sentence. (Note also *They have so/too not done their homework* vs. **They have not so/too done their homework*.)

1. Externalization.

The Y-model had syntax branching off in symmetric fashion to PF and LF. In “Overt vs. Covert Movement”, I in effect took this example of symmetry to be mistaken, by taking LF-movement not to exist. Chomsky’s ‘externalization’ continues this ‘branching off from syntax’ tradition, as do other proposals for ‘post-syntactic’ operations.

Chomsky’s thinking that linear order is not part of core syntax seems to be driven in part by his belief that linear order plays no role in semantic interpretation, as opposed to hierarchical structure.

In the opposite spirit, Kayne (1994, 131) had “From this perspective, linear order turns out to be more fundamental to syntax than is normally thought. As a result of the LCA, the property of antisymmetry that a linear ordering has is inherited by the hierarchical structure.”

2. Pronouns and their non-c-commanding antecedents.

There is some evidence, though, that linear order is relevant to antecedent-pronoun relations. Consider the following, from the interaction between pronouns and their antecedents.

English readily allows both:

(1) The fact that John is here today means that he’s well again.

(2) The fact that he’s here today means that John is well again.

Both (1) and (2) have the property that in them neither *John* nor *he* c-commands the other, with English thereby giving the impression that in such non-c-command configurations anything goes. This impression may have fed into Lasnik’s (1976) claim that pronouns could in general freely take antecedents subject only to (what later came to be called) conditions B and C of the binding theory.⁴

Yet it appears that English is not representative, insofar as various other languages allow sentences like (2) less readily than English, or not at all. For example, Michel DeGraff (p.c.) has told me that in Haitian Creole “backward pronominalization”⁵ of the sort seen in (2) is systematically impossible. In addition, Huang (1998, sect. 5.5.2) indicates that Chinese has much less backward pronominalization than English. And Craig (1977, 150) in her grammar of Jacaltepec says that Jacaltepec has no backward pronominalization at all. Strikingly (since Danish is a Germanic language like English), Allan et al.’s (1995, 473) grammar of Danish says that Danish has either none or at least much less backward pronominalization than English (cf. Thráinsson et al. (2004, 331) on Faroese, another Germanic language).⁶ Finally, Jayaseelan (1991, 76) says

⁴On the idea that neither Condition B nor Condition C is primitive, see Kayne (2002).

⁵This term goes back to the 1960s; cf. Langacker (1969).

⁶Cf. also Antonyuk and Bailyn (2008) on Russian.

about Malayalam that some speakers of Malayalam allow no backward pronominalization at all.

In contrast to these sometimes severe limitations on backward pronominalization found in some languages, I don't know of any language that in a systematic way prohibits forward pronominalization of the sort seen in (1). There thus seems to be a precedence-based asymmetry of considerable importance concerning antecedent-pronoun relations in contexts of non-c-command, of a sort that would be unexpected in a symmetric syntactic universe, and unexpected if interpretation were truly blind to linear order.

A proposal to account for this asymmetry was made in Kayne (2002). Oversimplifying a bit, the proposal was that the language faculty requires that the antecedent of a pronoun must precede that pronoun at some point in the derivation.⁷

If so, then in (2), there must be a stage in the derivation at which *John* precedes *he* (as the result of leftward, sideward movement of *John*⁸), and there must subsequently apply in the derivation of (2) a further movement operation (or more than one) that accounts for *he* subsequently coming to precede *John*.

The expectation now, from a comparative syntax perspective, is that languages with no backward pronominalization will not have access to that type of derivation, in particular not to that further movement operation. Languages with some backward pronominalization (but less than English) will have only limited access to the relevant type of derivation.⁹

That the antecedent of a pronoun must precede that pronoun at some point in the derivation means, on the assumption that antecedent-pronoun relations are part of core syntax, that precedence, too, must be part of core syntax.

3. More on linear order as part of core syntax.

The conclusion just reached is at odds with the suggestion that Chomsky (2019; 2020) has made in recent work, when he says that linear order is not part of core syntax and that linear order comes into play only as the result of what he calls externalization. Let us now ask to what extent antisymmetry might or might not be compatible with Chomsky's view of externalization.

⁷Cf. Bruening (2014). Kayne (2002, section 7) explicitly took discourse to be a subtype of coordination (cf. Hoekstra (1999)), in order to integrate cases like:

i) John is famous. He's smart, too.

⁸Sideward movement of the type put forth by Bobaljik and Brown (1997) and Nunes (2001; 2004). On the formalization of sideward movement, cf. Collins and Stabler (2016, 50).

That sideward movement should only be leftward does not follow from the LCA; it recalls Abels and Neeleman (2006) taking what was a "theorem" in Kayne (1994) to the effect that movement is always leftward and elevating it to an "axiom." It also recalls Kayne (1994, 37) and especially Kayne (2011, section 4) on time slots.

⁹What kind(s) of parameter(s) underlie these differences remains to be determined.

There could be a link to (only apparently rightward) heavy-NP-shift (of one object past another).

Antisymmetry as in Kayne (1994) claims that linear order is completely dependent on structure; in particular for every projection, the mapping to linear order must invariably yield Spec-Head-Complement order, whatever the type of projection. There is no optionality of the head-parameter sort. Antisymmetry also requires that there be only one Spec per projection, in a way that has fed into cartography work, as exemplified by Rizzi (1997) and Cinque (1999).

My impression is that from Chomsky's externalization perspective one could, if one agrees that S-H-C is correct, in fact incorporate antisymmetry into a mapping from core syntax to PF.

Chomsky (2020) takes the view that linear/temporal order is not at all part of core syntax.¹⁰ In part he does so on the basis of the point that differences in linear order do not feed differences in semantic interpretation, which depends only on structure (and on lexical items and features).

But the force of this point rests on the assumption, denied by antisymmetry, that linear order can vary independently of structure (as it could in the head parameter tradition). If, on the other hand, linear order is fully determined by hierarchical structure, then there is no reason to expect it to be able to make its own independent contribution to semantic interpretation. (Though differences in linear order could well correlate with differences in interpretation if the associated differences in hierarchical structure do.) In which case linear order can, as far as interpretation is concerned, perfectly well be part of core syntax.)

4. Even more on linear order as part of core syntax.

Another reason why Chomsky has taken linear order not to be part of core syntax has to do with examples that show that internal merge cannot take the linearly closest auxiliary in English subject-aux inversion.¹¹ Thus starting from:

(3) Somebody who is in Paris is on the phone.
one cannot conceivably derive:

(4) *Is somebody who in Paris is on the phone?

The question arises, though, as to whether this strong prohibition might (or might not) derive from independent structural factors, e.g. from the general impossibility of extraction from within a subject phrase, or from within a relative clause. Relevant, needless to say, is the fact that extraction from within a subject phrase or from within a relative clause does not always yield a violation as strong as that of (4). For example, to my ear the following instance of extraction from within a subject is less sharply deviant than is (4):

(5) ???He's somebody who close friends of generally like linguistics a lot.
The difference in degree of acceptability, as compared with (4), is even stronger with parasitic gaps:

(6) ?He's somebody who close friends of generally admire.

One can, on the other hand, reach a violation as sharp as that in (4) if, instead of extracting an argument from within a subject, as in (5) and (6), one tries to extract a non-argument such as an adverb. Thus, starting from:

¹⁰Cf. Chomsky (1995, 340), going back to Chomsky (1970).

¹¹Indirectly relevant here is the question whether Internal Merge is triggered; for Chomsky (2019, 268) it is not.

(7) Somebody who was speaking loudly left very suddenly.
it is sharply impossible to derive:

(8) *How loudly did somebody who was speaking leave very suddenly?
and similarly even for parasitic gaps (where the notion of 'linearly closest' is not obviously relevant). For example, starting from:

(9) Somebody who was behaving badly was near somebody else who was behaving badly.

one cannot reach:¹²

(10) *How badly was somebody who was behaving near somebody else who was behaving?

Of importance now is the fact that a parasitic gap counterpart of (4) also remains very strongly deviant:¹³

(11) *Is somebody who in Paris on the phone?
even though 'linearly closest' is not relevant to parasitic-gap examples.

The same set of points can be made with regard to extraction from within a relative clause. As Chung and McCloskey (1983, 708) noted, English does allow such extraction in some cases, e.g.:¹⁴

(12) That's one trick that I've known a lot of people who've been taken in by.
However, this example, and all the others they cite (as being "either well formed or much more acceptable than one would expect"), are instances of the extraction of an argument.

Examples similar to any of theirs, but involving the extraction of an adverb, are invariably sharply deviant, as far as I can see. For example, alongside:

(13) We've known a lot of people who word their messages carefully.
there is no:

(14) *How carefully have you known a lot of people who word their messages.

The strong deviance of (8), (10), (11) and (14) is independent of any notion of linear closeness. Consequently, the strong deviance of (4), if assimilable to that of (8), (10), (11) and (14) in terms of structural constraints on the extraction of non-arguments, can be understood in a way that is fully compatible with linear order being part of core syntax.

5. Counting is not part of syntax.

Chomsky (2020) also mentions an aspect of work by Moro et al. (2003) showing that Broca's area activation does not take place when subjects are presented with an 'unreal' language in which, for example, negation would systematically be the third word in a sentence. This is a telling point, but I don't think it bears directly on the question whether linear order is part of core syntax. Rather, what it shows is that the language

¹²The same holds of Chomsky's (2020; 2022, 352) example:

i) Carefully the guy who fixed the car packed his tools.

The adverb *carefully* cannot originate within/be extracted from the relative clause embedded within the subject, whether there is in addition a parasitic gap or not.

¹³Cf. the fact, discussed by Källgren and Prince (1989, 55), that finite V-movement out of a relative clause island in Yiddish is impossible, even though it leaves a copy.

¹⁴Cf. Taraldsen (1981) on Norwegian, as well as Bolinger (1972).

faculty doesn't count numerically. Clearly the language faculty doesn't count words in linear order.

But it crucially also doesn't numerically count structural notions such as depth of embedding. As far as I know, no syntactic operation takes, or could possibly take, as its goal a phrase that would be exactly three nodes down from the (node immediately dominating the) probe. Nor could any syntactic operation search for a phrase that is the third closest, structurally speaking.

Another type of counting that is unattested and I would guess unattestable would be (in a way that could conceivably have been useful) for a language to give an early indication, in complemetizer-like fashion, of how many words a(n embedded) sentence contains, i.e. no language has anything like:

(15) *He thinks three Mary is smart.

(16) *He thinks four Mary is very smart.

That numerical counting is not in the above cases countenanced by the language faculty is itself something that needs to be understood (in particular against the background of Chomsky's (2019, (text to) note 27) suggestion that arithmetic is "the simplest case of language"),¹⁵ but its unavailability clearly does not imply that linear order is not part of core syntax. (Cf. the fact that counting of the sort discussed in not found in phonology, either.)

6. Core syntax and semantic interpretation.

I have taken the position that linear order doesn't need to directly feed into semantic interpretation, since linear order is antisymmetrically determined by hierarchical structure, which does feed into it. Yet linear order is part of core syntax. Let me take all this to be part of a broader question, namely whether or not semantic interpretation must take into account all aspects of core syntax.¹⁶

This broader question ties into one raised by Chomsky et al. (2019), who state that "Whether...semantically vacuous scrambling, extraposition, clitic movement etc., ...reflect narrow-syntactic computations or are part of the mapping to PHON...is an open question".¹⁷

¹⁵Languages can use numerals to count in certain cases, such as:

i) Mary, John and Susan have all three/*four of them voted for you.

¹⁶If semantic interpretation ignores some aspects of core syntax, that will have something in common with phonetic interpretation not requiring that all aspects of phonological structure have a phonetic counterpart, e.g. deleted/silent phonological segments, with one example being abbreviated words, as in *abs* (for *abdominal* (*muscle*)s), whose lax vowel (in my English) plausibly reflects the presence of silent DOMINAL at the relevant point in the derivation, contrasting phonetically thereby with *tabs*.

¹⁷Relevant here is Chomsky's (1995, 151) Full Interpretation principle, whose interaction with idioms calls for substantial clarification. Note also Matushansky's (2006, Appendix) arguments "against analyzing head movement as a phonological phenomenon"; and similarly, Hosono (2019). Also relevant here is Arano's (2022, sect. 4.2) recent argument that scrambling in Japanese must take place in narrow syntax; cf. Saito and Fukui (1998, 445) on heavy-NP-shift (and scrambling). As well as Cinque's (2023, 102) discussion of 'meaningless movements'.

Consider the following proposal:¹⁸

(17) Arguments invariably raise at least once.

If this is correct, then even arguments that appear in their canonical/neutral position in the language in question (including in VO languages) will necessarily have been moved there in what appears to be a semantically vacuous way (unless the semantics pays important attention to the link between even an argument in canonical position and its trace).

7. VO order in non-finite environments in Estonian and Finnish.

But before I go any further into the question of diagnosing core syntax, let me indulge in a digression concerning (17) and canonical order, via a striking observation due to Vilkuna (1998), who states that Estonian and Finnish have verb-final order more consistently in non-finite adjuncts than in non-finite complements. Put conversely, those two languages have VO order more consistently in non-finite complements than in non-finite adjuncts. Now Brattico and Huhmarniemi (2006, 10) take the basic word order of Finnish to be SVO.¹⁹ The question, then, is why the prevalence of this basic/canonical VO order should be at all sensitive, in non-finite contexts, to whether the non-finite embedding is a complement or an adjunct.

If Finnish VO directly reflected external merge positions, contrary to (17), this sensitivity would be very difficult to understand. Whereas if (17) is valid, the likelihood of an account is greater.

That Finnish supports (17) seems clear. As usual, details need to be filled in. What comes to mind is this: VO order in nonfinite clauses in Finnish (and Estonian) is derived via a notable instance of scrambling. More specifically, the O in such cases must be long-distance scrambled out of the nonfinite clause entirely, to a position high up in the matrix. Subsequent to this, remnant movement of the matrix VP (which will include the nonfinite clause) takes the matrix VP to a still higher position.²⁰ The reason that VO order in non-finite clauses is less readily available in adjuncts now reduces to the fact that extraction (of O) out of adjuncts is less readily available than is extraction out of complements.

8. Diagnosing core syntax.

Let us now return to the question of types of evidence that bear on being part of core syntax or not. I will touch on cases in which semantically vacuous syntactic operations can be diagnosed as being part of core syntax indirectly, through their interactions with other core syntactic operations. Take, for example, relative clause extraposition, which feels semantically neutral in pairs like:

(18) Somebody who I used to know in high school just walked in.

(19) Somebody just walked in who I used to know in high school.

¹⁸Cf. Kayne (2024).

¹⁹Cf. Holmberg (2016, 27).

²⁰Cf. the derivations given in Kayne (1998b) for the matrix scope negation reading in English sentences such as those given by Klima (1964), e.g.:

(i) John will force his daughter to marry no one.

See also Bayer (1993).

Yet there is a restriction that comes to light in the following pair:

(20) The only person who I liked in high school just walked in.

(21) *The only person just walked in who I liked in high school.

presumably due to a property of the scope of *only* that must surely be (keyed to) part of core syntax. Put another way, relative clause extraposition is not available in (21) for syntactic/semantic reasons that cannot possibly be part of PF, even if relative clause extraposition itself is in general interpretively neutral.

This point concerning relative clause extraposition harks back to Holmberg's (1986, sect. 6.2) argument that Scandinavian Object Shift is syntactic, i.e. part of core syntax, in current terms. This is so in particular, he argues,²¹ because Object Shift feeds topicalization, itself obviously part of core syntax.

In a partially similar way, consider clitic movement of the Romance type, which in simple cases seems to have no semantic effect. Yet we have contrasts in French of the following sort:²²

(22) Combien (*en) ont lu ton livre? ('how-many (of-them) have read your book')

(23) Combien tu *(en) a lus? ('how-many you (of-them) have read' = 'how many of them have you read')

When one fails to pronounce the noun that goes with *combien* ('how many'), the clitic *en* ('of them') is obligatorily present in the object case (23),²³ yet impossible in the subject case (22). The obligatoriness in (23) tracks the obligatoriness of *en* in:

(24) Tu *(en) a lu beacoup. ('you (of-them) have read many')

suggesting strongly that the movement of clitic *en* in (23) must be taking place prior to wh-movement. If so, then, since wh-movement is part of core syntax, so must be clitic movement (at least in this case), despite its semantic neutrality.

That clitic movement is part of core syntax is also suggested by its interaction with raising to subject position, in cases in French such as:²⁴

(25) Le premier chapitre semble en être intéressant. ('the first chapter seems of-it to-be interesting')

The clitic *en* ('of it') originates within the DP containing *le premier chapitre* ('the first chapter'). It clearly must, in (25), move to clitic position within the embedded infinitival phrase prior to the raising to matrix subject position of *le premier chapitre*. Since that raising is part of core syntax, so must the movement of *en* be, despite its (apparent) semantic neutrality.²⁵

Finally, take verb raising of the sort discussed in Pollock (1989), which as Chomsky (2020) emphasizes is also semantically neutral. Yet in Kayne (1991) I argued that Romance infinitival verb raising has an effect on whether or not PRO is admissible as the subject of a Romance infinitival *if*-clause. In some Romance languages it is, and in

²¹Cf. also Holmberg and Platzack (1995, chapter 6) and Holmberg (2005).

²²Cf. Kayne (1975, sects. 2.19, 4.3), Rizzi (1982, 148ff.) and Pollock (1998).

²³In a resumptive pronoun-like fashion - cf. Kayne (2022a).

²⁴Originally discussed by Ruwet (1972).

²⁵Relevant here is Uriagereka (2000) on the semantics of clitic doubling; also Déprez (1998) and Obenauer (1992) on semantic effects of past participle agreement in French; as well as Ikawa (2022) on how Agree feeds interpretation with Japanese honorifics.

others it is not, and this correlates with whether or not the Romance language in question moves its infinitives past the landing site of its object clitics.²⁶ If so, then Romance infinitival verb raising must, since it bears on the admissibility of controlled PRO, be part of core syntax, despite apparently being semantically neutral.

Recent corroboration of this correlation comes from Kato et al.'s (2023, 309) observation that European Portuguese prohibits clitic-infinitive order with its counterpart of 'if'+infinitive:

i) Não sabemos se {*lhe dizer /dizer-lhe} a verdade hoje ou amanhã. ('neg we-know if him to-tell/to-tell him the truth today or tomorrow')
despite allowing clitic-infinitive order elsewhere.

9. Linear order as temporal order, or, The integration of time into core syntax.

Returning to the question of 'linear order'/'temporal order', we now see that to say that linear order is part of core syntax is to say that temporal order is part of core syntax. Conversely, to say that linear order is not at all part of core syntax is to say that temporal order is not at all part of core syntax. Which is in turn to say that the language faculty has not integrated into its core any notion of time.

Such a(n arguably counter-biological) position, or close to it, was in effect taken by Chomsky (1995, 221), who said "If humans could communicate by telepathy, there would be no need for a phonological component, at least for the purposes of communication."

The question for future work (not limited to linguistics) is whether or not the human brain has or has not entirely disregarded time at the very core of one of its most important components, as well as to what extent the human brain integrates notions of time in other areas of cognition, broadly interpreted.

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²⁶This correlation fits in with Kato et al.'s (2023, 309) observation that European Portuguese prohibits clitic-infinitive order with its counterpart of 'if'+infinitive:

i) Não sabemos se {*lhe dizer /dizer-lhe} a verdade hoje ou amanhã. ('neg we-know if him to-tell/to-tell him the truth today or tomorrow')
despite allowing clitic-infinitive order elsewhere. Whether the proposal in Kayne (1991) can extend to comparable facts found in EP with certain prepositions remains to be worked out.

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